

Primal Dual Applied to Max Flow

max: v

$$\text{st: } Af + dv \leq 0, \quad d = \begin{pmatrix} -1 \\ +1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \begin{matrix} s \\ t \end{matrix}$$

$$f \leq b$$

$$-f \leq 0$$

We want to  combinationize the RHS "b"
hence treat this as the dual problem.

Max Flow as the Dual



Dual Restricted Primal

Need: Dual feasible solution f

J = index of the dual constraints that are tight.

DRP: max: v

st: $Af + dv \leq 0$ // flow conservation

$f \leq 0$ for rows where $f = b$ in D (b is given)

$-f \leq 0$ for rows where $f = 0$ in D

$$f \leq 1$$

$$v \leq 1$$

DRP Solution

If $v=0$: f was optimal. stop.

To set $v=1$: There must be a

path from s to t that uses

saturated arcs ($f_{ij}=b_{ij}$ in D) in backward
direction ($\bar{f} \leq 0$), and arcs with zero

flow ($f_{ij}=0$ in D) in forward direction ($-\bar{f} \leq 0$)

and other arcs in either direction.